**MultiModalR: Two Functions to Rule Them All—Automatic Bayesian Mixture Modeling for Multimodal Data**

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Abstract

Most mixture modeling tools require users to specify the number of components (*K*), a critical parameter that is often unknown in practice. Manual selection of *K* introduces subjectivity and error, especially for multimodal data with categorical grouping variables. We introduce *MultiModalR*, an R package that automatically estimates the number of components via density-based mode detection. The package implements efficient Markov Chain Monte Carlo sampling in optimized C++ and provides two complementary approaches: a fast, parallel stratified model for independent category fits, and a principled hierarchical model that shares information across categories with category-specific parameters. Benchmarked against established packages (*Rmixmod*, *mclust*, *bgmm*, *flexmix*) and the recent *BayesMultiMode*, *MultiModalR* achieves superior performance in both component detection and clustering quality. The hierarchical model attains perfect K-detection accuracy (100%) across all difficulty levels—including challenging overlapping datasets—while the stratified model maintains robust performance (53–100%). In a real-world ecological application using hyperspectral forest data, the hierarchical model achieves 96.1% mode plausibility, outperforming the stratified model (92.8%) and demonstrating practical utility. Unlike *BayesMultiMode*, *MultiModalR* provides per-observation component assignments with probability estimates, making it suitable for both exploratory mode analysis and supervised classification. By eliminating manual K-specification and reducing analysis time from hours of trial-and-error to seconds of automated computation, *MultiModalR* advances reproducible and objective mixture modeling for multimodal data in environmental and ecological sciences.

Keywords: Bayesian mixture modeling · Multimodal data · Mode detection · Markov Chain Monte Carlo · R package · Hierarchical modeling · Automatic K-selection · Hyperspectral remote sensing

## 1. Introduction

Identifying latent subpopulations within heterogeneous data is a fundamental challenge across the sciences. Finite mixture models are a cornerstone of model-based clustering, but a critical bottleneck for practitioners is specifying the number of components (*K*), which is often unknown. Mixture modeling is a cornerstone of model-based clustering and classification, enabling the identification of latent subpopulations within heterogeneous data (McLachlan and Peel, 2000). Bayesian and maximum likelihood (ML) mixture models - particularly Gaussian and multinomial - have been extensively studied by statistical researchers (Alamichel et al., 2024; Beraha et al., 2021; Bull et al., 2023; Grün and Malsiner-Walli, 2022; Haines, 2016; Kurita et al., 1992; Neal, 1992; Rezapour and Ksaibati, 2021; Van der Vaart, 1996). These are implemented in several widely used R packages, each with distinct methodological strengths.

For example, *Rmixmod* provides a comprehensive interface to the ML-based Mixmod library, supporting unsupervised, supervised, and semi-supervised classification with a rich set of covariance structures and estimation strategies (Lebret et al., 2015). *mclust* uses the Expectation-Maximization (EM) algorithm for model-based clustering and classification and employs Bayesian information criterion (BIC) for model selection (Scrucca et al., 2016). *flexmix* implements a flexible framework for finite mixtures of regression models and latent class analysis within an ML framework using EM algorithm (Leisch, 2004). *bgmm* extends Bayesian mixture modeling to partially supervised settings through belief-based and soft-label assignment for handling uncertain labels (Biecek et al., 2012).

Additionally, several specialized, less widely adopted tools address more targeted methodological gaps. For instance, *EMcluster* implements EM algorithms and efficient initialization methods for finite Gaussian mixture models with unstructured dispersion, supporting unsupervised and semi-supervised clustering along with utilities for model evaluation and visualization (Chen et al., 2025). *mixR* provides a unified EM-based interface for fitting and comparing finite mixture models across multiple distributions - including Gaussian, Weibull, Gamma, and

Log-normal—for both raw and binned data (Yu, 2022). It features computationally efficient EM-based estimation via Rcpp and incorporates the bootstrapping likelihood ratio test (bLRT) (Feng and McCulloch, 1996; Yu and Harvill, 2019) alongside BIC for robust model selection.

Despite their sophistication, these tools share a fundamental constraint: the number of mixture components, denoted as *K* in this study, must be specified a priori or selected through repeated model-fitting and criterion comparison (e.g., using BIC or bLRT). This process is not only computationally demanding but also subjective, as researchers must manually inspect and compare multiple candidate models. In practice, this often leads to overfitting, underfitting, or irreproducible results. While BIC-based methods (as in *mclust* and *Rmixmod*) automate model selection to some extent, they still require fitting multiple models across a range of *K* values—a prohibitive task for large and high-dimensional datasets. Bayesian approaches like *bgmm*, though powerful in handling label uncertainty, do not automate the detection of *K*, leaving a critical gap in the modeling pipeline.

Alternative Bayesian approaches do exist for avoiding manual specification of *K*. Dirichlet process mixture models (DPMMs) assume an infinite number of components, with the data determining a finite subset (Ferguson, 1973; Valle et al., 2022). Reversible jump MCMC (RJMCMC) allows transdimensional moves between different values of *K* during sampling (Green, 1995; Richardson and Green, 1997). These methods are powerful and general, applicable to a wide range of mixture models beyond the Gaussian case. However, they are computationally intensive, require careful prior specification, and their implementation can be challenging for practitioners who use large, high-dimensional datasets.

Most recently, Basturk et al. (2026) introduced *BayesMultiMode*, an R package for Bayesian mode inference in univariate distributions. *BayesMultiMode* estimates the posterior probability of the number of modes and their locations after fitting a sparse finite mixture model. While it provides rigorous uncertainty quantification for mode counts and locations, it does not provide per-observation component assignments—a critical feature for classification tasks. *MultiModalR* addresses this complementary gap by providing not only automated K-detection but also probabilistic individual-level assignments, making it suitable for both exploratory mode analysis and supervised classification.

This paper introduces *MultiModalR*, a novel Bayesian mixture modeling framework that integrates automatic component detection with a complete, production-ready workflow. Our contributions are threefold:

(i) Methodological innovation: We introduce a fully automated two-stage algorithm that first detects mixture components via height-aware kernel density mode detection, and then performs Bayesian inference via Metropolis-Hastings within Gibbs sampling (MH-within-Gibbs) (Gamerman and Lopes, 2006; Tierney, 1994). This two-stage approach determines *K* from the data, removing the need for manual specification or iterative model comparison. Importantly, the framework includes both a **stratified** model (independent fits per category) and a **hierarchical** model (shared components with category-specific parameters), providing flexibility for different data structures.

(ii) Software implementation: We provide an open-source R package (*MultiModalR*) that supports parallel processing, built-in validation, MCMC diagnostics (Effective Sample Size, Gelman-Rubin convergence checks), and seamless integration into analytical pipelines.

(iii) Empirical validation: Through extensive simulations and a real-world ecological application using remotely sensed hyperspectral data, we demonstrate that *MultiModalR* outperforms existing methods in both accuracy and efficiency, while providing rigorous convergence diagnostics.

By eliminating the need for manual K-specification and providing per-observation assignments, *MultiModalR* advances the reproducibility, objectivity, and scalability of mixture modeling for multimodal data in environmental and ecological sciences.

## 2. MultiModalR

### 2.1. Height-aware mode detection

Our mode detection algorithm employs kernel density estimation. Users can select one of four bandwidth selection methods: SJ-dpi (Sheather and Jones, 1991), nrd (Silverman, 2018), bcv (Scott and Terrell, 1987), and ucv (Bowman, 1984; Rudemo, 1982). Significant peaks are identified as local maxima exceeding a threshold relative to the mean density. Peaks are then grouped using a height-aware algorithm: peaks are processed in descending order of density height, with each high peak forming a new group that absorbs any unprocessed peaks within a specified radius (Fig. 1). This approach ensures that prominent modes anchor their own groups while merging minor nearby peaks.

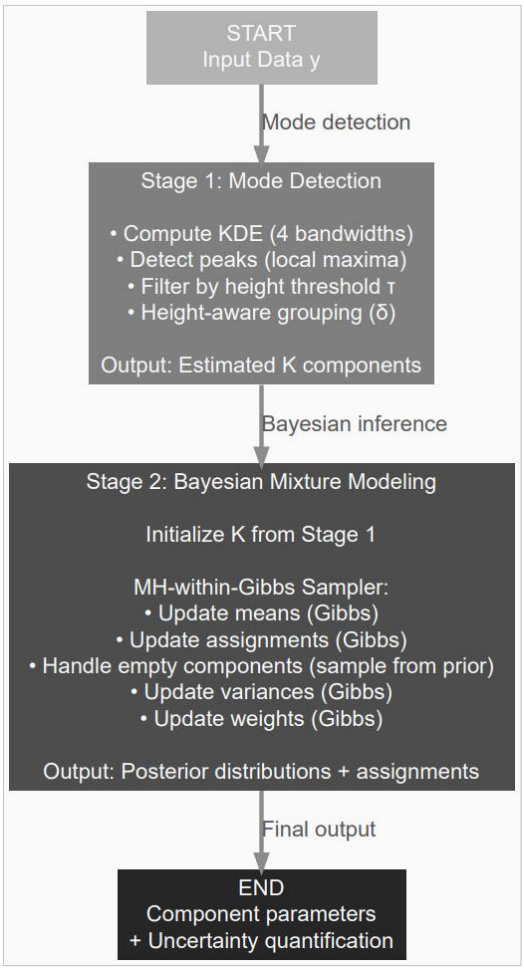



Figure 1. Flowchart of our two-staged algorithm, stage one: height-aware mode detection and stage two: Bayesian mixture modeling.

The mathematical description of the algorithm is as follows:

Let $y = \{y_1, \ldots, y_n\}$ be univariate observations. We compute kernel density estimates using a user-selected bandwidth method $m \in \{\text{SJ, nrd, bcv, ucv}\}$. For method $m$, the density estimate is:

$$\hat{f}^{(m)}(x) = \frac{1}{nh_m} \sum_{i=1}^{n} K\left(\frac{x-y_i}{h_m}\right), \qquad \text{(Eq. 1)}$$

where $h_m$ is the bandwidth for method $m$, and $K(\cdot)$ is the Gaussian kernel.

Let $P^{(m)}$ denote the set of significant peaks detected using bandwidth method $m$. For the density estimate $\hat{f}^{(m)}(x)$ evaluated on a grid $\{x_1, \ldots, x_{M_{grid}}\}$ of $M_{grid}$ equally spaced points, we identify local maxima as:

$$P^{(m)} = \{(x_j, h_j) : h_j = \hat{f}^{(m)}(x_j), \hat{f}^{(m)}(x_j) > \hat{f}^{(m)}(x_{j-1}) \text{ and } \hat{f}^{(m)}(x_j) > \hat{f}^{(m)}(x_{j+1})\}. \qquad \text{(Eq. 2)}$$

We retain only peaks exceeding a relative threshold $\tau > 0$ (Fig. 1):

$$h_j > \tau \cdot \bar{f}^{(m)}, \text{ with } \bar{f}^{(m)} = \frac{1}{M_{grid}} \sum_{j=1}^{M_{grid}} \hat{f}^{(m)}(x_j). \qquad \text{(Eq. 3)}$$

The height-aware grouping proceeds as follows:

1. Sort by height: Arrange peaks in descending order of density height:

$$P^{(m)}_{sorted} = \{(x_{(1)}, h_{(1)}), \ldots, (x_{(L)}, h_{(L)})\}, \quad h_{(1)} \geq h_{(2)} \geq \ldots h_{(L)}, \qquad \text{(Eq. 4)}$$

Where L = $|P^{(m)}|$ is the total number of peaks detected.

2. Initialize: Marks all peaks as unprocessed. Set group index $g = 1$.

3. Group peaks: For $i = 1$ to $L$.:

If peak $i$ is unprocessed:

- Assign it to group $G_g$
- Mark it as processed
- For all unprocessed peaks $j > i$:
  - If $|x_{(j)} - x_{(i)}| \leq \delta$ (merging radius), assign peak $j$ to group $G_g$ and mark as processed
- Increment group index: $g \leftarrow g + 1$.

4. Final modes: For each group $G_g$, compute the mean location of its constituent peaks:

$$\tilde{x}_g = \frac{1}{|G_g|} \sum_{(x,h) \in G_g} x, \quad \text{(Eq. 5)}$$

where $|G_g|$ denotes the number of peaks in group g.

The estimated number of components is $K = |\{\tilde{x}_1, \ldots, \tilde{x}_K\}|$.

Default parameters are $\delta = 0.1$ (merging radius called *within*) and sj_*adjust* = 0.5 (bandwidth multiplier). These defaults provided robust performance across most of our validation studies (Section 3.2), but parameters can be adjusted by users for specific applications. The detected modes serve as initial component means for both the stratified and hierarchical mixture models described below.

### 2.2. MH-within-Gibbs

Following automatic component detection (Section 2.1), *MultiModalR* provides two complementary modeling approaches: a stratified model and a hierarchical model. Both use the MH-within-Gibbs sampler described below, but differ in how they handle the categorical variable.

Stratified Model

The stratified model fits an independent Gaussian mixture for each level of the categorical variable (*varCLASS*). For each category $c$ with $K_c$ components, the generative model is:

$$y_i \mid (C_i = c, z_i = \text{k}) \sim N(\mu_c, k, \sigma^2_{c,k}), \; \text{P}(z_i = k \mid C_i = c) = \pi_{c,k}, \quad \text{(Eq. 6)}$$

where $z_i$ indicates component assignment and $C_i$ is the category of observation $i$. and $k = 1, \ldots, K_c$. This approach is computationally efficient (parallelizable) and suitable for exploratory analysis when categories are expected to have independent structures.

Hierarchical Model

The hierarchical model treats the categorical variable as a covariate, sharing the number of components $K$ across all categories while allowing category-specific means, variances, and weights. For each observation $i$ with category $c$, the generative model is:

$$y_i \mid (C_i = c) \sim \Sigma_{k=1}^{K} \pi_{c,k} \, N(\mu_{c,k}, \sigma^2_{c,k}), \; \Sigma_{k=1}^{K} \pi_{c,k} = 1, \quad \text{(Eq. 7)}$$

where $\mu_{c,k}$, $\sigma^2_{c,k}$ and $\pi_{c,k}$ are the category-specific mean, variance, and mixing weight for component $k$. This formulation borrows information across categories—the shared $K$ stabilizes estimates, particularly for small categories—while maintaining flexibility through category-specific parameters.

MCMC sampling

For both models, we specify conjugate priors. For the stratified model, for each component $k$ in category $c$:

$$\mu_{c,k} = N(m_{c,k}, 1), \quad (\pi_{c,1}, \ldots, \pi_{c,K_c}) \sim \textit{Dirichlet}(\alpha, \ldots, \alpha), \quad \sigma^2_{c,k} \sim \textit{InverseGamma}(\alpha_0, \beta_0),$$

where $m_{c,k}$ is set to the $k$-th detected mode for category $c$ from Section 2.1.

For the hierarchical model, the priors are analogous but with shared $K$:

$$\mu_{c,k} = N(m_{c,k}, 1), \quad (\pi_{c,1},\ldots, \pi_{c,K}) \sim Dirichlet(\alpha,\ldots,\alpha), \qquad \sigma^2_{c,k} \sim InverseGamma(\alpha_0, \beta_0).$$

The MH-within-Gibbs sampler iterates through the following steps:

1. Update means $\mu_{c,k}$: For the stratified model, the full conditional is Gaussian and we use a Gibbs step. For the hierarchical model, the full conditional is not conjugate due to category-specific weights; we use a random-walk Metropolis-Hastings step with proposal standard deviation *proposal_sd*.

2. Update assignments $z_i$ via Gibbs sampling (Fig. 1):

$$\mathbb{p}(z_i = k \mid \cdots) \propto \mu_{c,k} \cdot N(y_i \mid \mu_{c,k}, \sigma^2_{c,k}) \cdot \exp(-\gamma \cdot n^{(-i)}_{c,k}), \qquad \text{(Eq. 8)}$$

where $n^{(-i)}_{c,k}$ is the number of observations in category *c* currently assigned to component *k*, excluding observation *i*, and $\gamma > 0$ is a small repulsion parameter (default: $\gamma = 0.01$). This term penalizes components that would become too large, encouraging a more balanced distribution across components and preventing the collapse of the mixture into a single dominant component. This repulsion mechanism stabilizes the Markov chain and improves mixing, particularly in multimodal posteriors with overlapping components.

3. Handle empty components: If a component has no observations assigned, we sample its parameters from their priors, allowing it to potentially acquire observations in subsequent iterations without breaking the Markov chain's reversibility.

4. Update weights $\pi_{c,k}$ via Gibbs sampling from the Dirichlet posterior (Fig. 1). For the stratified model:

$$(\pi_{c,1},\ldots, \pi_{c,K_c}) \mid \cdots \sim Dirichlet(\alpha + n_{c,1}, \ldots, \alpha + n_{c,K_c}). \qquad \text{(Eq. 9)}$$

For the hierarchical model:

$$(\pi_{c,1},\ldots, \pi_{c,K}) \mid \cdots \sim Dirichlet(\alpha + n_{c,1}, \ldots, \alpha + n_{c,K}), \qquad \text{(Eq. 10)}$$

where $n_{c,K}$ is the number of observations in category *c* currently assigned to component *k*.

5. Update variances $\sigma^2_{c,k}$ via Gibbs sampling (Fig. 1).

At each MCMC iteration, sample each $\sigma^2_k$ from its full conditional InverseGamma posterior:

$$\sigma^2_{c,k} \mid \cdots \sim InverseGamma(\alpha_0 + \tfrac{n_{c,k}}{2}, \beta_0 + \tfrac{1}{2}\textstyle\sum_{i:z_i=k, C_i=c}(y_i - \mu_{c,k})^2). \qquad \text{(Eq. 11)}$$

Label switching and identifiability: In standard Bayesian mixture modeling, label switching occurs when component indices permute during MCMC sampling without changing the likelihood (Stephens, 2000). To address this, we enforce an ordering constraint $\mu_{c,1}<\mu_{c,2}<\ldots<\mu_{c,K}$ within each category after each MCMC iteration (for the hierarchical model; analogously for the stratified model). This ensures that component labels remain interpretable throughout the chain.

MCMC convergence diagnostics: To assess convergence, we provide two diagnostics in the package. First, the Effective Sample Size (ESS) measures the number of effectively independent samples from the posterior, accounting for autocorrelation. Second, the Gelman-Rubin potential scale reduction factor ($\hat{R}$) compares within-chain and between-chain variances across multiple independent chains; values below 1.1 indicate convergence (Gelman and Rubin, 1992). These diagnostics are automatically available for the hierarchical model.

The algorithm is implemented in optimized C++ via RcppArmadillo for computational efficiency (Eddelbuettel et al., 2025, 2024). Default hyperparameter is $\alpha = 5$ (Dirichlet prior) and, for the hierarchical model, $\alpha_0 = 3.0$ and $\beta_0 = 2.0$ (Inverse Gamma prior). This value was empirically tuned to provide stable mixing across a range of simulation scenarios.

### 2.3. Two functions to rule them all

Users are provided with two main functions that perform the complete two-stage process described previously. The stratified approach is executed by the *fuss_PARALLEL_mcmc()* function—where *fuss* is the Hungarian translation for 'run'—and the hierarchical approach is executed by the *fuss_COVARIATE_mcmc()* function, as

illustrated in Fig. 2.

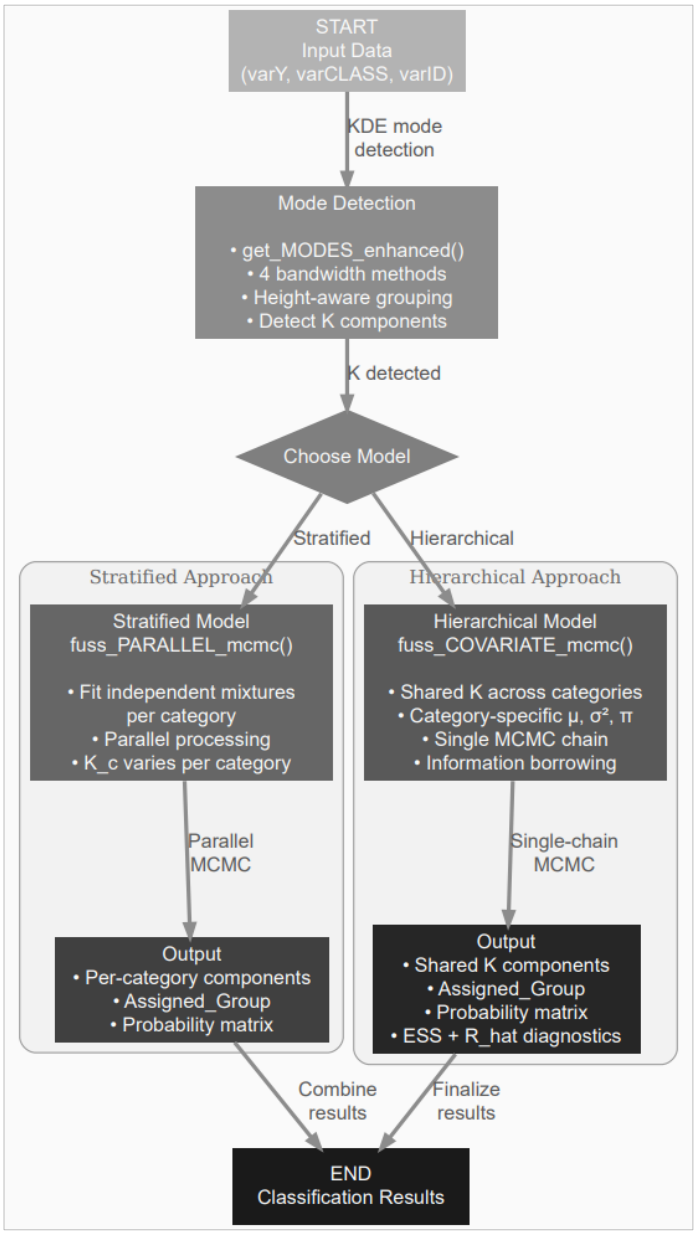



Figure 2. Flowchart of the integrated analysis pipeline in *MultiModalR*. Following mode detection, users can choose between the stratified model (*fuss_PARALLEL_mcmc()*) for independent category fits (parallel processing) or the hierarchical model (*fuss_COVARIATE_mcmc()*) for shared components with category-specific parameters (single chain with convergence diagnostics).

Both functions follow the same core pipeline: data input → kernel density estimation → mode detection → Bayesian MCMC → output generation. However, they differ in how the categorical variable is handled: *fuss_PARALLEL_mcmc()* fits independent mixtures for each category (stratified), while *fuss_COVARIATE_mcmc()* fits a unified model with shared components and category-specific parameters (hierarchical).

Regarding both functions' parameterization, four data-dependent parameters (*data*, *varCLASS*, *varY*, and *varID*) must be provided by the user. Throughout this study, these parameters were set consistently based on the input data. An in-depth overview detailing all parameters in *MultiModalR* including their implications, default values, and permissible ranges, is presented in Table 1.

Table 1. Parameter characteristics of *fuss_PARALLEL_mcmc()* and *fuss_COVARIATE_mcmc()*.

| Name | Default value | Implication | Range or values |
|---|---|---|---|
| *data** | - | Input data frame | [-] |
| *varCLASS** | - | Category variable name | [-] |
| *varY** | - | Value variable name | [-] |
| *varID** | - | ID variable name | [-] |
| *method* | sj-dpi | Density estimator method | [sj-dpi, nrd, bcv, ucv] |
| *K*** | NULL | Number of mixture components | [≥2] |
| *within**** | 1.0 | Height range parameter (δ) for grouping modes | [≥ 0] |
| *maxNGROUP* | 5 | Maximum number of groups | [≥ 2] |
| *out_dir***** | NULL | Output directory for csv files (NULL returns combined data frame) | [-] |
| *n_workers***** | 3 | Number of parallel workers | [≥ 2] |
| *n_iter* | NULL | NULL means 6000 for 'metropolis' and 3000 for 'dirichlet' | [≥ 1000] |
| *burnin* | NULL | NULL means 2000 for 'metropolis' and 1000 for 'dirichlet' | [≥ 500] |
| *proposal_sd* | 0.15 | Proposal standard deviation for component means | [> 0] |
| *sj_adjust**** | 0.5 | Adjustment factor for bandwidth selector | [> 0] |
| *mcmc_method* | metropolis | MCMC method: MH-within-Gibbs or Dirichlet | [metropolis, dirichlet] |
| *dirichlet_alpha* | 2.0 | Dirichlet concentration parameter | [> 0] |

* input data-dependent parameters
** *fuss_COVARIATE_mcmc()* parameter only
*** Advised tuning parameters for users seeking to refine mode detection or density estimation
**** *fuss_PARALLEL_mcmc()* parameter only

Two advised tuning parameters (*within* and *sj_adjust*) were adjusted from their default values for specific analyses. For the real-world data (Section 3.1.2), *within* was set to 0.1 because the range of the value variable was constrained between 0 and 1. For the CSSA (Section 3.2.1), *sj_adjust* was set to 1.0 (no adjustment to the default SJ-dpi bandwidth). Default values were used for all other analyses, including the KDR benchmark (Section 3.2.2). All analyses used the MH-within-Gibbs sampler.

To illustrate the package's workflow, the following minimal examples demonstrate fitting both models to the package-provided dataset:

```
# Load package and data
library(MultiModalR)
df <- MultiModalR::multimodal_dummy

# ---- Stratified model (parallel) ----
results_strat <- fuss_PARALLEL_mcmc(
```

```
  data = df,
  varCLASS = "Category",
  varY = "Value",
  varID = "ID"
)

# ---- Hierarchical model (single chain) ----
results_hier <- fuss_COVARIATE_mcmc(
  data = df,
  varY = "Value",
  varCLASS = "Category",
  K = 3,                # Auto-detected if NULL
  n_iter = 10000,
  burnin = 2000
)

# View results
head(results_strat)
head(results_hier$prob_matrix)
```

| | ID | y | Group_1 | Group_2 | Group_3 | Group_4 |
|---|---|---|---|---|---|---|
| AA.1 | 1 | 6.091588 | 0.9572772 | 0.04272027 | 2.580177e-06 | NA |
| AA.2 | 2 | 6.073984 | 0.9598540 | 0.04014370 | 2.286025e-06 | NA |
| AA.3 | 3 | 6.526434 | 0.7743712 | 0.22557185 | 5.694471e-05 | NA |
| AA.4 | 4 | 5.719141 | 0.9863035 | 0.01369629 | 2.082639e-07 | NA |
| AA.5 | 5 | 5.603510 | 0.9895619 | 0.01043798 | 9.701669e-08 | NA |
| AA.6 | 6 | 6.428236 | 0.8448357 | 0.15513599 | 2.829545e-05 | NA |

| | Group_5 | Assigned_Group | Min_Assigned | Max_Assigned |
|---|---|---|---|---|
| AA.1 | NA | 1 | 5.313286 | 6.57687 |
| AA.2 | NA | 1 | 5.313286 | 6.57687 |
| AA.3 | NA | 1 | 5.313286 | 6.57687 |
| AA.4 | NA | 1 | 5.313286 | 6.57687 |
| AA.5 | NA | 1 | 5.313286 | 6.57687 |
| AA.6 | NA | 1 | 5.313286 | 6.57687 |

| | Mean_Assigned | Mode_Assigned | Main_Class |
|---|---|---|---|
| AA.1 | 6.008625 | 6.103026 | AA |
| AA.2 | 6.008625 | 6.103026 | AA |
| AA.3 | 6.008625 | 6.103026 | AA |
| AA.4 | 6.008625 | 6.103026 | AA |
| AA.5 | 6.008625 | 6.103026 | AA |
| AA.6 | 6.008625 | 6.103026 | AA |

```
# Inspect the assigned groups
table(results$assigned_group)
```

The output for both models includes the original data, posterior probabilities for each component (Group_1, Group_2, ...), the assigned group (Assigned_Group), and category-level statistics. For the hierarchical model, additional MCMC diagnostics are available via *check_convergence()* and *gelman_rubin()* functions.

## 3. Validation: Materials and methods

### 3.1. Data sets

#### 3.1.1. Generated data

Two synthetic datasets were generated, each comprising nine distinct categories and three latent subpopulations. The first dataset, named *AAtoII* after its category labels (AA, BB, …, II), was designed with slight overlap between the latent distributions. Each observation was assigned its true subpopulation label to enable validation.

Consequently, the *AAtoII* data exhibits three relatively evenly distributed peaks (Fig. 3). This dataset was used for the CSS analysis.

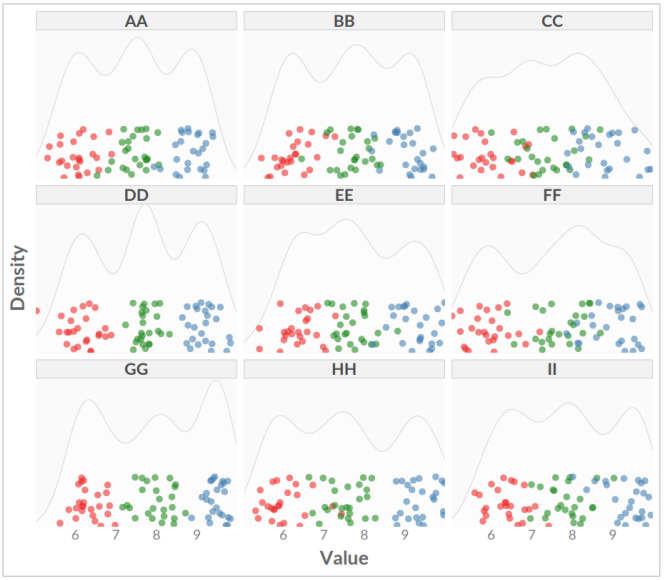



Figure 3. The generated *AAtoII* data with nine categories, each containing three latent subpopulations. The dataset comprises 675 records in total, with 25 samples generated per subpopulation.

The second dataset was constructed to systematically vary the overlap intensity between subpopulations. Its categories were explicitly labeled to reflect this design: AA ~ Easy, BB ~ Easy, CC ~ Easy, DD ~ Medium, EE ~ Medium, FF ~ Medium, GG ~ Hard, HH ~ Hard, and II ~ Hard (Fig. 4). This dataset is hereafter referred to as *EMH* (named for its Easy, Medium, and Hard overlap levels). As with *AAtoII*, each observation was assigned its true subpopulation label for validation. For the KDR assessment, the *EMH* data generation process was repeated 10 times using distinct random seeds (123, 124, ..., 132) to create replicated datasets. As intended, the number and clarity of the density peaks in each replication are directly influenced by the degree of overlap, providing a graded challenge for mode detection.

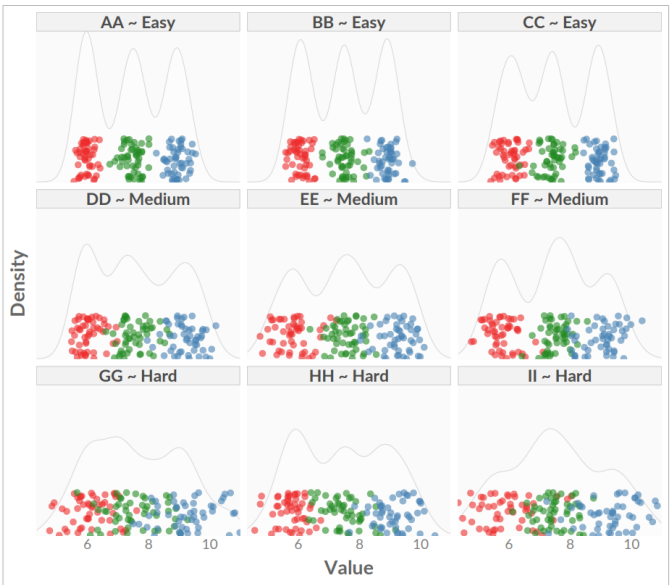



Figure 4. Example of *EMH* data generated using random seed 124. The data comprises nine major categories, each containing three distinct latent subpopulations. The full dataset consists of 1350 total records, with each subpopulation represented by 50 samples.

### 3.1.2. Ecological data

To evaluate the *MultiModalR* package on complex, real-world data, we used the 10-fold cross-validated classification accuracies from a companion study on tree species classification (Abriha et al., 2026) as our input dataset. The source data consisted of hyperspectral imagery from three distinct forests in Hungary—S (Ságvár), M (Mecsek), and G (Gemenc) (Fig. 5)—each containing a variety of tree species.

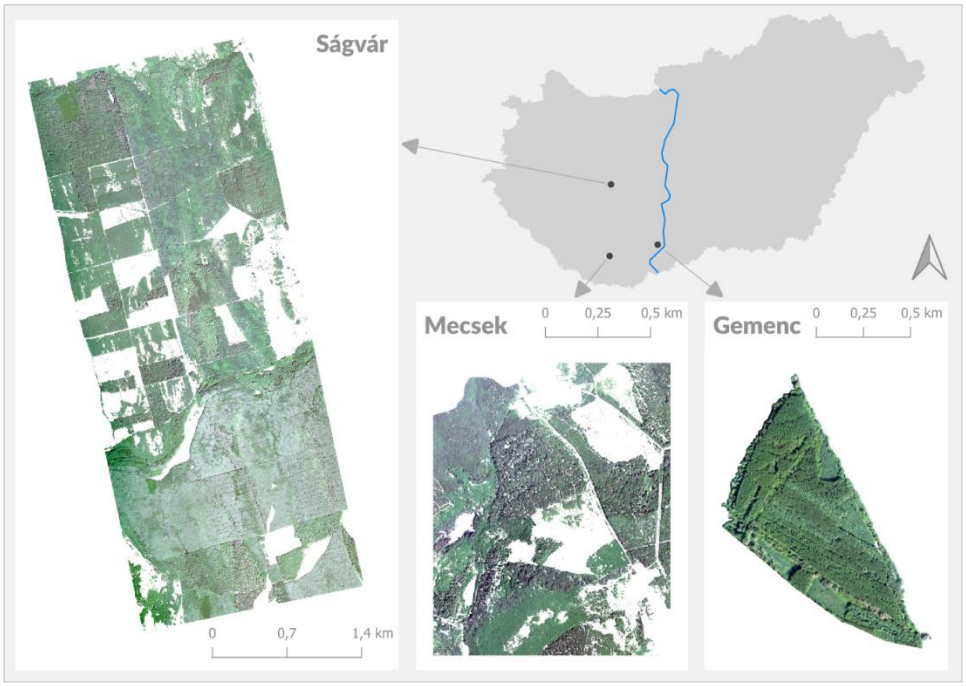



Figure 5. Locations of the three Hungarian forests. The Danube river is shown with a blue line, while white spaces in the maps indicate areas without forest cover.

In that study, a Random Forest classifier (Breiman, 2001) was applied to the original imagery and to 81 versions filtered with a Savitzky-Golay (SG) smoother (Savitzky and Golay, 1964). The SG parameters were varied across a comprehensive grid (window sizes: 3-19; polynomial order: 1-18), creating 81 unique image variants per forest. For our multimodal analysis, the overall classification accuracy distribution for each forest—derived from these 81 parameter sets and averaged across 10 cross-validation folds—serves as the empirical input (Fig. 6). These accuracy vectors form a high-dimensional, heterogeneous dataset. We hypothesize that the observed multimodality primarily originates from the distinct spectral signatures of different tree species, which constitute the latent subpopulations. The realistic challenge for the mode detection pipeline arises precisely because these spectral signatures can overlap between species, creating complex, non-ideal multimodal distributions.

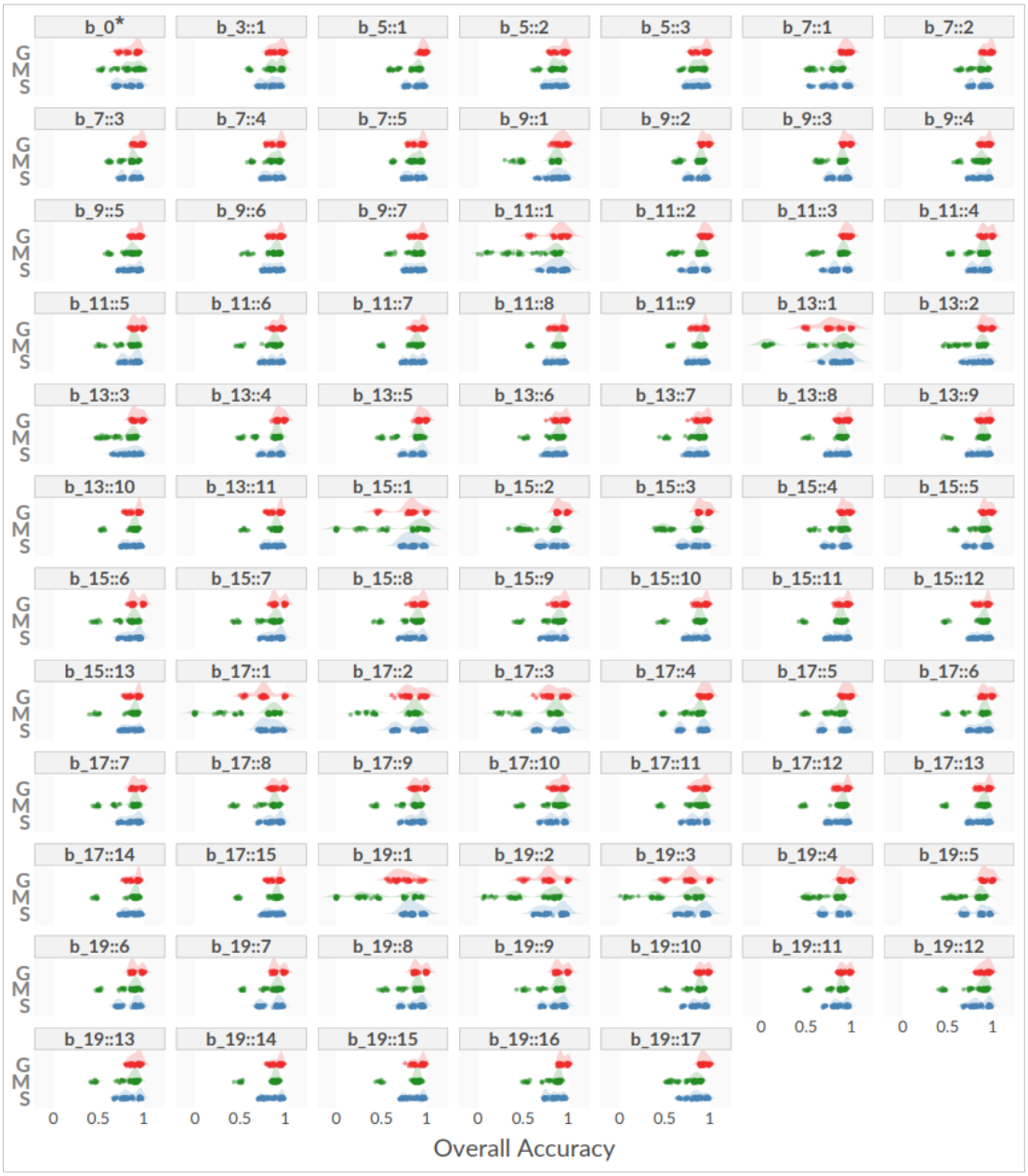



Figure 6. Averaged 10-fold cross-validated classification overall accuracy for each Hungarian forest—Ságvár (S), Mecsek (M), and Gemenc (G)—across all 82 image variants (81 distinct Savitzky-Golay smoothed versions and the original, unsmoothed data marked with an asterisk, *). Each panel corresponds to a unique combination of Savitzky-Golay parameters (Window size::Order) and shows the performance distribution for that variant via raw points and density estimates. The number of data points per forest across all variants is 80 for S, 70 for M, and 50 for G.

## 3.2. Methods

### 3.2.1. Certainty-Separation-Stability Analysis (CSSA)

The CSSA framework was applied to the stratified model (*fuss_PARALLEL_mcmc()*) on the *AAtoII* dataset (Section 3.1.1). Certainty quantifies the confidence of individual assignments. We calculated for each observation $i$ its maximum posterior assignment probability, $p_i^{max} = max_k p_{ik}$, where $p_{ik}$ is the posterior probability that $i$ belongs to component $k$, provided by the sampler. The distribution of $p_i^{max}$ values, summarized by its mean $\bar{p}^{max}$, reflects overall assignment certainty. High average certainty indicates that the model assigns most observations to a single component with high confidence, a common diagnostic in model-based clustering (Fraley and Raftery, 2002; Scrucca et al., 2016). For reporting, certainty is expressed as a percentage: Certainty (%) = $\bar{p}^{max} \times 100$. Following established convention (Fraley and Raftery, 2002; Scrucca et al., 2016), we consider mean certainty above 80% as indicative of well-separated components.

Separation measures the statistical distinctness between the assigned subpopulations. For each pair of inferred components (*k*, *l*), we compute a simplified version of Cohen's *d* using Eq. 12 (Cohen, 2013):

$$d_{kl} = \frac{|\bar{y}_k - \bar{y}_l|}{\sqrt{(s_k^2 + s_l^2)/2}}, \quad \text{(Eq. 12)}$$

where $\bar{y}_k$ and $s_k^2$ are the mean and variance of the data points assigned to component *k*. The minimum pairwise Cohen's *d* across all component pairs is reported as the separation metric, with $d_{min} > 1$ suggesting moderate separation and $d_{min} > 3$ indicating very strong separation. To convert separation to a percentage scale comparable with the other CSSA metrics, we apply a saturating transformation: Separation (%) = min(100, $\frac{d_{min}}{5} \times 100$). This maps $d_{min} = 5$ (or greater) to 100%, reflecting essentially complete separation. Thus, a score > 60% corresponds to very strong separation ($d_{min} > 3$).

Stability assesses the reproducibility of the assignments across multiple independent MCMC runs. We executed the sampler $R = 20$ times with different random seeds. For each pair of runs (*r*, *s*), we calculated the pairwise agreement proportion $A_{rs}$, defined as the fraction of observations assigned to the same component in both runs. The overall stability metric is the mean across all unique run pairs:

$$\bar{A} = \frac{2}{R(R-1)} \sum_{r<s} A_{rs}. \quad \text{(Eq. 13)}$$

High mean pairwise agreement indicates that the multimodal structure is consistently recovered regardless of initialization, a key criterion for validating cluster solutions (Hennig, 2007; Lange et al., 2004). Stability is reported as a percentage: Stability (%) = $\bar{A} \times 100$. Stable reproducibility is generally considered above 95% (Hennig, 2007).

#### 3.2.2. Benchmarking K-Detection and Robustness (KDR)

For a comprehensive performance comparison, seven established R packages for mixture modeling and clustering were initially identified as benchmarks for *MultiModalR*: *Rmixmod*, *BayesMultiMode*, *mclust*, *bgmm*, *flexmix*, *EMCluster*, and *mixR*. However, the model selection performance of *EMCluster* and *mixR* was found to be inadequate for our synthetic benchmark data, leading to their exclusion from the formal comparative analysis. An overview of the considered packages and their inclusion status is provided in Table 2.

Table 2. Mixture-modeling R packages considered for benchmarking against *MultiModalR*.

| R package | K-detection | Clustering | Selected for our study |
|---|---|---|---|
| *Rmixmod* | BIC, ICL, NEC | EM and SEM | Yes |
| *BayesMultiMode* | PMD | SFM | Yes |
| *mclust* | BIC | EM | Yes |
| *bgmm* | BIC | Bayesian EM | Yes |
| *flexmix* | BIC, ICL | EM | Yes |
| *EMCluster* | BIC | EM | No |
| *mixR* | BIC and bLRT | EM | No |

(Abbreviations: BIC = Bayesian Information Criterion; ICL = Integrated Completed Likelihood; NEC = Normalized Entropy Criterion; PMD = Posterior Mode Probability; SFM = Sparse Finite Mixture; EM = Expectation-Maximization; SEM = Stochastic EM.; bLRT = bootstrapping Likelihood Ratio Test)

For the KDR, the selected packages (Table 2) and *MultiModalR*—in both its stratified (*fuss_PARALLEL_mcmc()*) and hierarchical (*fuss_COVARIATE_mcmc()*) forms—were evaluated on the *EMH* dataset (Section 3.1.1). The evaluation was conducted over 10 independent simulation iterations using three core metrics: the accuracy of

component number detection, the robustness of the recovered assignment, and computational efficiency. Note that *BayesMultiMode* provides K-detection via posterior mode probabilities but does not produce per-observation assignments, so ARI cannot be computed for this method.

K-detection accuracy was calculated as the proportion of simulation runs in which a package's model selection criterion (we used BIC) correctly identified the true, pre-specified number of latent components ($K_{true}$). This metric, reported as a percentage, provides a direct measure of each algorithm's reliability in uncovering the correct multimodal structure.

The quality and stability of the final cluster assignment were quantified using the Adjusted Rand Index (ARI) (Hubert and Arabie, 1985). The ARI is a chance-corrected measure of the agreement between two data partitions - here, the estimated assignment and the known simulation ground truth. It is computed from the contingency table of the two partitions and adjusts for the expected agreement due to random chance, formulated as:

$$\text{ARI} = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}, \quad \text{(Eq. 14)}$$

where $n_{ij}$ represents the contingency table between true and estimated assignments, and $a_i$ and $b_j$ are the row and column sums. An ARI of 1 indicates perfect recovery of the true partition, 0 indicates agreement no better than random, and negative values indicate worse-than-random performance. For intuitive comparison with the K-detection metric, the ARI is also reported as a percentage ARI (%) = ARI × 100. ARI is considered as a standard metric for validating the robustness of clustering solutions (Steinley, 2004; Wagner and Wagner, 2007).

Execution time (in seconds) was recorded for each package to assess computational efficiency. Timing was performed using R's *system.time()* function, capturing the total elapsed time for the complete model fitting and selection process on the dataset.

### 3.2.3. Sample-To-Sample Assessment (S2SA)

The true robustness of the latent subpopulation detection was assessed at the individual record level on the *AAtoII* and *EMH* datasets (Section 3.1.1). Both datasets contain a unique identifier (ID), enabling direct comparison between the true latent label and the estimated assignment. For the *EMH* dataset, this assessment was performed across the same 10 independent simulation iterations used in the KDR (Section 3.2.2) using both the stratified (*fuss_PARALLEL_mcmc()*) and hierarchical (*fuss_COVARIATE_mcmc()*) models.

Match rate quantifies the S2SA predictive accuracy. For each main data category (e.g., AA, BB, … II), the estimated component labels were first aligned to the true labels by matching the order of their component means, resolving the common label-switching ambiguity inherent to mixture models. The match rate for a category was then calculated as the proportion of records where the aligned estimated label matched the true latent subpopulation:

$$\text{Match Rate} = \frac{1}{N}\sum_{i=1}^{N} I(TrueLabel_i = AlignedEstimatedLabel_i), \quad \text{(Eq. 15)}$$

where $I(\cdot)$ is the indicator function and $N$ is the number of records in the category. The overall robustness for a dataset is reported as the average match rate across all categories, expressed as a percentage: Match Rate (%) = Match Rate × 100.

### 3.2.4. Mode Plausibility Assessment (MPA)

The fidelity of multimodal detection was assessed at the individual data-point level using the real-world hyperspectral-derived dataset (Section 3.1.2). The MPA tests whether each observation is assigned to a plausible mode—that is, whether its assigned mode corresponds to any significant peak in the observed data distribution.

For each combination of SG parameters (82 groups), we first identified all significant modes by applying kernel density estimation and detecting local maxima. For each data point, the estimated mode value (*ModeAssigned*) from *MultiModalR* was compared to all observed modes for its group. A point was considered plausibly assigned if its assigned value fell within a predefined tolerance ($\tau$ = 0.05) of any observed mode in that group.

Group accuracy was calculated as the proportion of plausibly assigned points. Overall point-to-mode plausibility was reported as the mean accuracy across all groups, expressed as a percentage: Plausibility (%) = Accuracy × 100. Formally, for group $g$ with $N_g$ points and observed modes $M_g$, the match status for point *i* is:

$$MatchStatus_i = I(\min_{m \in M_g} |ModeAssigned_i - m| \leq \tau), \quad \text{(Eq. 16)}$$

where $I(\cdot)$ is the indicator function, and $\tau = 0.05$. The group plausibility is then:

$$Plausibility_g = \frac{1}{N_g} \sum_{i=1}^{N_g} MatchStatus_i. \quad \text{(Eq. 17)}$$

The MPA was performed using both the stratified (*fuss_PARALLEL_mcmc()*) and hierarchical (*fuss_COVARIATE_mcmc()*) models to compare their performance on real-world data.

## 4. Results

### 4.1. CSSA

For the nine categories in the dataset (*AAtoII*), a robust overall Certainty level was achieved. Seven of the nine categories scored above the 80% threshold defined in Section 3.2.1 (Fig. 7). Categories BB and EE scored marginally below this threshold, at 79% and 78% respectively. The minimal deviation of these values from the 80% benchmark supports the overall robustness of MH-within-Gibbs' (i.e., *MultiModalR*'s) Certainty metric.

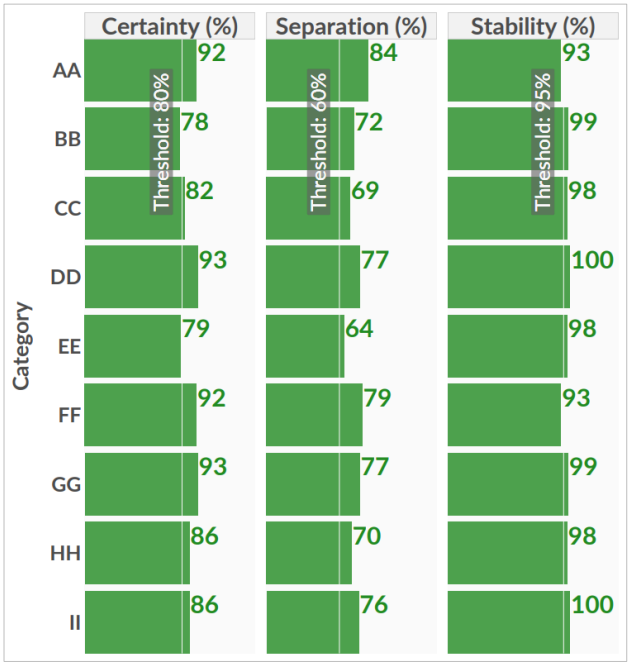



Figure 7. Certainty-Separation-Stability Assessment (CSSA) results for the *AAtoII* dataset. Metrics are expressed as percentages (horizontal bars); corresponding thresholds (Section 3.2.1) are indicated by the white vertical lines.

All categories also achieved superior performance in Separation (Fig. 7), with scores exceeding the 60% threshold that denotes excellent class separability (as described in Section 3.2.1). These results confirm the ability of our multimodal framework to achieve strong separation between latent subpopulations across the dataset.

For Stability, overall robustness was maintained. Categories AA and FF both achieved 93%, remaining very close to the 95% threshold (Fig. 7). The remaining categories met or exceeded this threshold, indicating stable reproducibility of the MH-within-Gibbs sampler.

### 4.2. KDR

In K-detection, both the stratified and hierarchical versions of *MultiModalR* demonstrated clear superiority at the Medium and Hard difficulty levels (Fig. 8). At the Medium level, the stratified model achieved a mean K-detection accuracy of 87% (SD = 23%), while the hierarchical model achieved 100% (SD = 0%). The third-best performer, *Rmixmod*, scored 73% (SD = 21%). At the Hard difficulty level, *MultiModalR* again led with a mean accuracy of 53% (SD = 23%) for the stratified model and 100% for the hierarchical model. At this level *Rmixmod* achieved 37% (SD = 25%). This consistent advantage confirms that *MultiModalR* delivers the most robust K-detection performance across challenging scenarios.

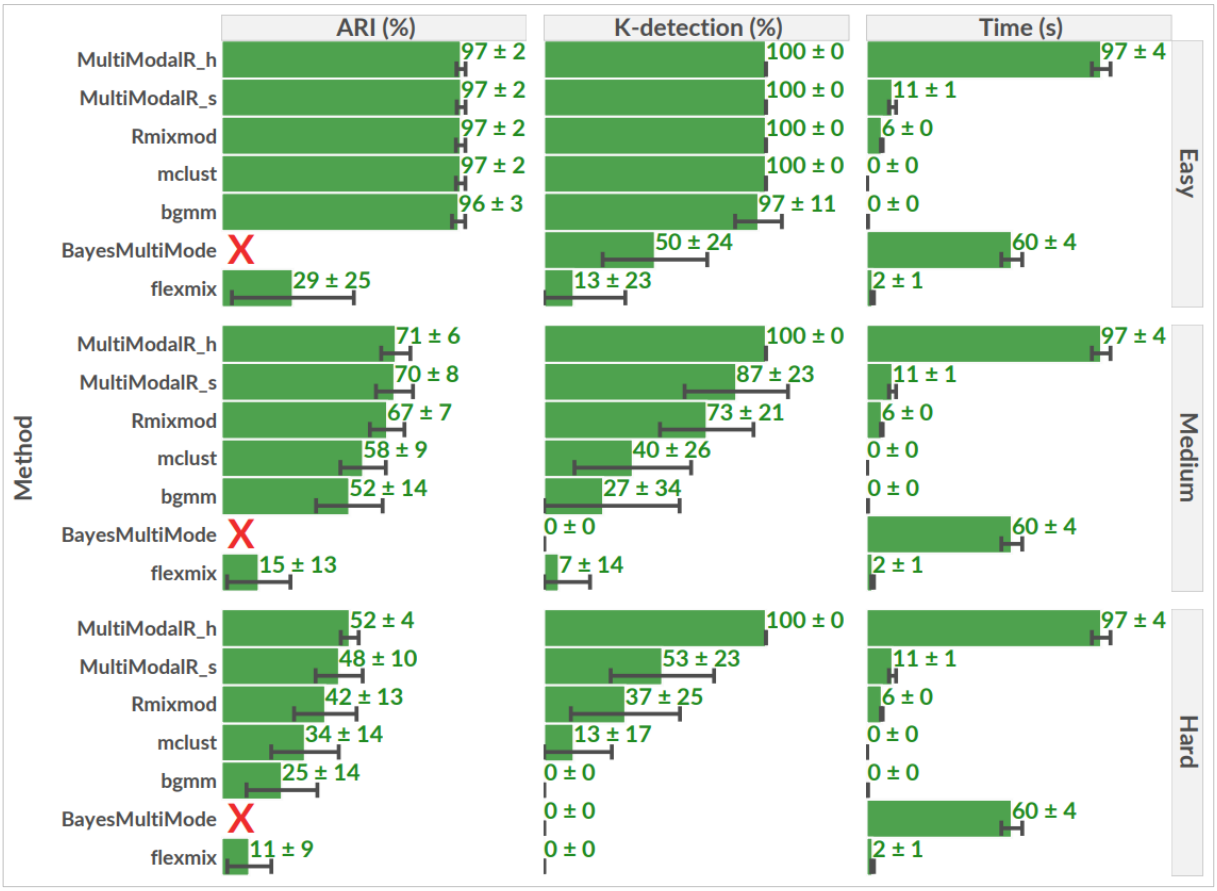



Figure 8. Benchmark results for K-detection and robustness assessment across Easy, Medium, and Hard difficulty levels. Metrics shown are Adjusted Rand Index (ARI, %), K-detection accuracy (%), and processing time (seconds). Competing tools include *MultiModalR_h (hierarchical)*, *MultiModalR_s (stratified)*, *Rmixmod*, *mclust*, *bgmm*, *BayesMultiMode* (ARI not available as it does not provide per-observation assignments), and *flexmix*. Performance scores are displayed as horizontal bars; standard deviations are indicated by dark grey error bars (capped to 0).

Consistently superior clustering quality was reflected in the ARI, where *MultiModalR* (38–99%) outperformed all competitors (2–99%) across all difficulty levels (Fig. 8). This advantage was most pronounced under challenging conditions: at the Hard difficulty level, *MultiModalR* maintained 38–58% ARI accuracy while competitors dropped to 2–55%. These results confirm that the method not only detects the correct number of components but also accurately recovers the underlying data partition.

Regarding computational efficiency *MultiModalR* and *BayesMultiMode* were the slowest (Fig. 8). Their mean processing times were 60 and 97 seconds, respectively, reflecting the computational cost associated with their statistical procedures.

### 4.3. S2SA

*MultiModalR* demonstrated excellent S2SA match rates across all validation scenarios. For the *AAtoII* dataset, all nine categories achieved match rates equal to or above 87% (Fig. 9).

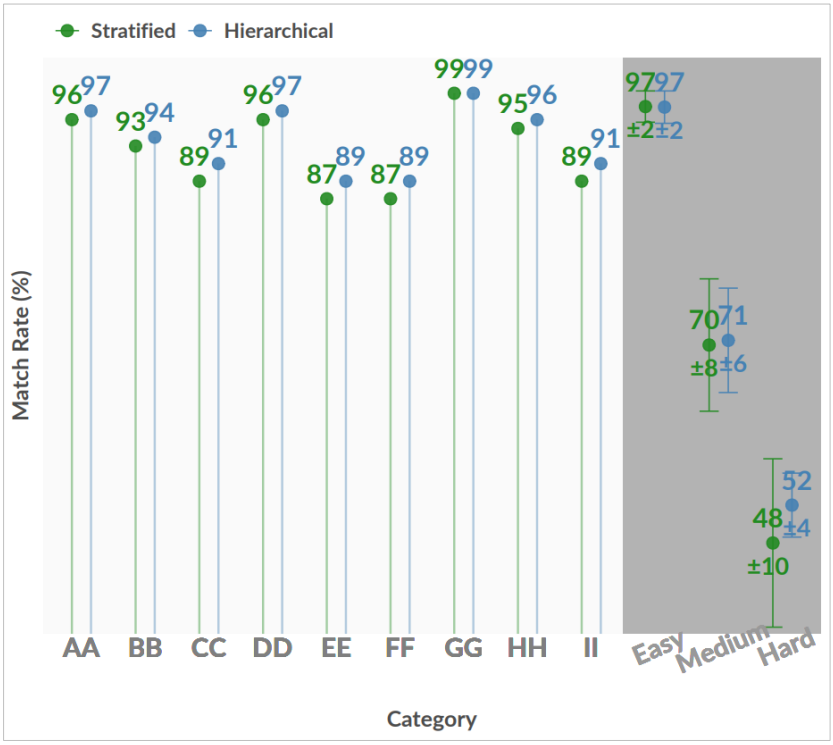



Figure 9. Sample-To-Sample Assessment (S2SA) results. Match rates for the nine categories in the *AAtoII* dataset (light grey background) and aggregated match rates for the nine categories across Easy, Medium, and Hard difficulty levels in the *EMH* benchmark dataset (dark grey background). Values of mean Match Rate (%) are shown as green (stratified) and blue (hierarchical) points with numerical labels above (mean) and below (standard deviation) for the *EMH* dataset; error bars indicate ±1 SD.

Performance remained strong on the more challenging *EMH* dataset. At the Medium difficulty level, the hierarchical model maintained a very good mean match rate of 71% (SD = 6%) (Fig. 9). At the Hard level—featuring overlapping and mixed subpopulations—the minimum mean match rate was 38%, while the highest 58%—both evidenced by the stratified model. These results confirm that *MultiModalR* retains robust subpopulation-prediction capabilities even under high-difficulty conditions.

### 4.4. MPA

The stratified model achieved an overall plausibility of 92.8%, with 213 out of 246 groups (86.6%) attaining perfect plausibility (100%) (Fig. 10). The hierarchical model substantially improved these results, achieving an overall plausibility of 96.1%—an absolute improvement of +3.3 percentage points—with perfect plausibility attained in 234 out of 246 groups (95.1%), representing an additional 21 perfect groups compared to the stratified model.




Figure 10. Difference in MPA accuracy between hierarchical and stratified models (Hierarchical − Stratified) across all Savitzky-Golay parameter combinations. Blue bars indicate hierarchical improvement; green bars indicate stratified improvement. Dark blue bars denote groups where the hierarchical model achieved perfect plausibility; the single light blue bar indicates improvement without perfection. The hierarchical model achieves an overall improvement of +3.3 percentage points (92.8% → 96.1%), increasing perfect groups from 213 to 234 out of 246. Groups are arranged by Savitzky-Golay parameter and forest type (G = Gemenc, M = Mecsek, S = Ságvár).

Figure 10 reveals that the hierarchical model consistently outperforms the stratified model across the vast majority of groups. The improvement is particularly evident for the Mecsek (M) and Ságvár (S) forests, where the hierarchical model achieved near-perfect plausibility in previously problematic parameter combinations. Critically, in groups where the hierarchical model achieved perfect plausibility, it not only detected the correct modes but also assigned the vast majority of individual points to their corresponding high-plausibility modes—demonstrating reliable point-level assignment, not just mode detection. The hierarchical model's superior performance stems from its ability to share information across categories, stabilizing estimates particularly for groups with limited data or ambiguous modal structures.

## 5. Discussion

### 5.1. Principal findings

Our validation demonstrates that *MultiModalR* successfully automates the critical step of component number selection in mixture modeling. The principal findings are:

(i) Superior K-detection accuracy. *MultiModalR* consistently identified the correct number of latent components across varying difficulty levels, achieving 30–100% accuracy compared to 0–100% for competing methods (Fig. 8). This performance gap was most pronounced under challenging conditions with overlapping components (Medium and Hard levels), where traditional methods relying on BIC-based model selection faltered. Critically, the hierarchical model achieved perfect K-detection (100%) across all difficulty levels, demonstrating the power of information sharing across categories.

(ii) Robust component recovery. The high Certainty (>78%), Separation (all categories >60%), and Stability (most categories ≥95%) metrics (Fig. 7) confirm that the algorithm not only detects the correct number of components but also yields statistically distinct and reproducible clusters. The strong S2SA match rates (>87% for *AAtoII*,

>48% hierarchical performance at Hard difficulty; Fig. 9) further validate accurate individual-level assignment. The hierarchical model consistently outperformed the stratified model, with match rates of 71% (SD = 6%) at Medium difficulty and up to 56% at Hard difficulty.

(iii) Real-world applicability. On complex ecological hyperspectral data, the hierarchical model achieved an overall MPA plausibility of 96.1%, compared to 92.8% for the stratified model—an absolute improvement of +3.3 percentage points. Perfect plausibility was attained in 234 out of 246 groups (95.1%), representing an additional 21 perfect groups compared to the stratified model (Fig. 10). This demonstrates that the hierarchical model transitions effectively from controlled simulations to noisy, derived real-world data, while the stratified model remains a strong baseline.

These results indicate that the two-stage approach—density-based mode detection followed by Bayesian inference—effectively eliminates the subjectivity of manual K-selection while maintaining robust statistical performance. The success of *MultiModalR* suggests that decoupling component detection from parametric fitting provides a more stable foundation for mixture modeling, particularly when latent distributions overlap. From a practical standpoint, this automation reduces analysis time from hours of manual trial-and-error to seconds, making advanced mixture modeling accessible to a broader range of researchers.

### 5.2. Methodological advances and trade-offs

The performance of *MultiModalR* can be attributed to its foundational design choices, which also delineate its scope and trade-offs relative to existing tools.

The density-first advantage. The key innovation is decoupling component detection from model fitting. Traditional methods (e.g., *Rmixmod*, *mclust*) must fit multiple models for different $K$ values and rely on information criteria (BIC, ICL) for selection. In multimodal data with overlapping components, these criteria often fail, leading to under/over-fitting—as evidenced by the poor K-detection accuracy of benchmarks at Medium and Hard difficulty levels (Fig. 8). In contrast, *MultiModalR*'s kernel density mode detection identifies candidate $K$ directly from the data's empirical distribution. This approach is inherently more robust to overlap because it responds to peaks in the data density itself, not just likelihood improvements in a parametric mode.

Stability through Bayesian repulsion. The MH-within-Gibbs sampler incorporates a repulsion mechanism (Eq. 8, $\gamma > 0$) that discourages empty components and stabilizes assignments by penalizing components that would become too large. This, combined with the handling of empty components (Step 3, Section 2.2), ensures the Markov chain explores plausible partitions anchored around the detected modes, explaining the high Stability and Certainty metrics (Fig. 7). The conjugate priors for variances and weights allow for efficient Gibbs updates, while Metropolis-Hastings steps for means provide the flexibility needed for multimodal posteriors.

Acknowledged trade-off. This robustness comes at computational cost. *MultiModalR* was slower (mean ~11s for stratified, ~97s for hierarchical) than some EM-based algorithms (Fig. 8). We argue this is a justifiable trade-off: the alternative—manually fitting and comparing 5–10 models per dataset—is more time-consuming and introduces subjectivity. *MultiModalR*'s one-function-pipeline (*fuss_PARALLEL_mcmc()* for stratified analysis and *fuss_COVARIATE_mcmc()* for hierarchical modeling) ultimately reduces total analysis time from hours of trial-and-error to seconds of automated computation, a clear net gain for most applied users.

### 5.3. Positioning within the mixture modeling ecosystem

Our work bridges a critical gap identified in the review of existing software (Section 1). Packages like *Rmixmod* and *mclust* excel at fitting mixtures given $K$; *bgmm* extends this to uncertain labels; *flexmix* offers unparalleled regression flexibility. However, none automate the initial determination of $K$ for multimodal categorical data in a way that is both computationally efficient and accessible to applied researchers. *MultiModalR* thus completes the ecosystem by providing a principled, automated entry point that removes the most subjective step in the workflow.

Comparison with *BayesMultiMode*. The recently introduced *BayesMultiMode* (Basturk et al., 2026) provides Bayesian mode inference for univariate distributions using sparse finite mixtures. While both packages address mode detection, *MultiModalR* offers distinct advantages: (i) it provides per-observation component assignments and probabilities, enabling classification tasks; (ii) it offers both stratified and hierarchical modeling approaches; (iii) it provides comprehensive MCMC diagnostics (ESS, $\hat{R}$) for convergence assessment; and (iv) its C++ implementation offers computational efficiency. In our KDR assessment, *MultiModalR* outperformed *BayesMultiMode* in K-detection accuracy across all difficulty levels, highlighting the benefit of the density-first approach for component detection.

Implications for reproducible research. Manual $K$ selection is a 'researcher degree of freedom' that can

lead to inconsistent findings across studies. By automating this choice via a transparent, algorithmically defined process (height-aware mode detection), *MultiModalR* standardizes the first and most consequential step in mixture analysis. The package's tuning parameters (*within*, *sj_adjust*) offer expert users the flexibility to refine results without reverting to full manual specification, striking a balance between automation and control.

Translational impact. The successful application to hyperspectral ecological data (Sections 3.1.2 and 4.4) demonstrates translational impact. The latent subpopulations, likely corresponding to tree species with distinct spectral signatures, were identified automatically from a derived accuracy metric. This suggests wide applicability in environmental sciences, genomics, psychometrics, and any field where categorical data exhibits complex, overlapping multimodal structures. By making advanced mixture modeling accessible through the intuitive *fuss_PARALLEL_mcmc()* and *fuss_COVARIATE_mcmc()* functions, *MultiModalR* lowers the barrier to entry for researchers who need to uncover latent structure but lack the time or expertise for manual model selection.

### 5.4. Limitations and directions for future research

While *MultiModalR* represents a substantial advance in automating mixture modeling for applied researchers, several limitations outline a clear path for future development. These limitations do not diminish the package's current utility but highlight opportunities for methodological expansion and refinement.

Univariate mode detection. The current implementation identifies modes independently for each level of the categorical variable (*varCLASS*). For truly multivariate multimodal data, this approach may miss components that are apparent only in feature space. Future work should develop genuine multivariate mode detection algorithms to broaden applicability to high-dimensional datasets.

Distributional scope. The validation focused on Gaussian (continuous) mixtures. Although the package also implements a Dirichlet Multinomial sampler for categorical data, extending the framework to other distributional families (e.g., Poisson, Beta, Weibull) would further increase its utility across diverse applied fields.

Hierarchical model refinement. The hierarchical model currently shares a single $K$ across all categories. While this assumption works well when categories have similar component structures, future work could allow per-category $K$ while still borrowing strength. Additionally, computational optimization of the C++ implementation could reduce the runtime overhead of the hierarchical approach.

## 6. Conclusions

*MultiModalR* provides a complete solution to a fundamental bottleneck in mixture modeling: the manual specification of component numbers ($K$). By integrating automated, density-based mode detection with optimized Bayesian inference, the package transforms a subjective, iterative process into an objective, one-step analytical workflow. The package offers two complementary approaches—a fast, parallel stratified model (*fuss_PARALLEL_mcmc()*) for exploratory analysis and a principled hierarchical model (*fuss_COVARIATE_mcmc()*) that shares information across categories for superior accuracy.

Our comprehensive validation establishes that *MultiModalR* outperforms established alternatives (*Rmixmod*, *mclust*, *bgmm*, *flexmix*) in K-detection accuracy, particularly for challenging datasets with overlapping components. The hierarchical model achieved perfect K-detection (100%) across all difficulty levels, while the stratified model maintained strong performance (87% at Medium, 53% at Hard). Both models demonstrated robust assignment quality (high CSSA metrics), excellent predictive match rates in simulation, and strong performance on complex real-world ecological data. Compared to *BayesMultiMode*, *MultiModalR* provides the additional benefit of per-observation assignments and probabilities, making it suitable for both exploratory mode analysis and supervised classification.

*MultiModalR* thus represents a paradigm shift toward objective, reproducible mixture modeling. By reducing analysis time from hours of manual trial-and-error to seconds of automated computation, it makes advanced latent class analysis accessible to a broader range of researchers. Future development will focus on extending the core methodology to multivariate mode detection and additional distributional families.

## 7. CrediT authorship contribution statement

**Gergő Diószegi**: Conceptualization, Methodology, Software, Visualization, Validation, Writing. **Dávid Abriha**: Ecological data Collection and Preparation. **Szilárd Szabó**: Editing.

## 8. Conflict of interest statement

None declared.

## 9. Acknowledgements

## 10. Data availability

*MultiModalR* package is available at https://github.com/DijoG/MultiModalR, and can be installed by executing the following line:

*devtools::install_github("DijoG/MultiModalR")*.

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